

Syed Shayan Mazahir

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EDUCATION

University of Waterloo

Expected June 2030

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, Java, Racket

Hardware & Embedded: Arduino, VEX Robotics Platform, Circuit Design, Voltage Regulation, Sensor Integration (Ultrasonic, IR, Gyroscopic), Motor Control Systems, PWM, PID Controllers

Software & Frameworks: Node.js, Express, Flask, Django, Git/GitHub, MongoDB, PostgreSQL, REST APIs

Areas of Interest: Embedded Systems, Robotics, Autonomous Navigation, Machine Learning, Natural Language Processing, Human-Computer Interaction

EXPERIENCE

Robotics Team Lead & Engineer

2024 – 2025

Louise Arbour Secondary School

- Won First Place in RobOlympics by leading 15-student team to design 4 autonomous robots; personally engineered voltage regulation circuits (5V to 15V series conversion), electrical schematics, and complete wiring harnesses for power distribution systems
- Achieved autonomous navigation by implementing A* pathfinding and PID control loops in C++; integrated ultrasonic/IR sensors, gyroscopes, and motor encoders using sensor fusion techniques for real-time positioning and obstacle avoidance
- Built modular robot platform with interchangeable hardware modules including servo-controlled radar, LCD display interface, and H-bridge motor drivers with PWM speed regulation; enabled rapid prototyping of competition strategies
- Trained 15 team members in embedded C++ programming, circuit debugging with oscilloscopes/multimeters, and VEX platform architecture; created technical documentation with annotated schematics and code repositories for knowledge transfer

Programming Hub Mentor

2024 – 2025

Louise Arbour Secondary School

- Improved student engagement by 60% by developing Flask web application with Google API integration serving 500+ students; conducted UX research and iterative testing to optimize platform features
- Mentored 20+ students in Python programming, algorithms, and data structures; tracked learning outcomes and performance data to iteratively refine curriculum and teaching approaches
- Secured student data by implementing SSL/HTTPS encryption and OAuth 2.0 authentication; ensured compliance with school privacy policies and security regulations

Aspect Systems

2022 – 2025

Community Operations & Infrastructure Lead

- Reduced incident response time from hours to minutes by building automated Python workflow systems for community moderation; implemented event-driven architecture with real-time alerting for scalable operations
- Contributed to C++-based anti-cheat system that analyzes player behavior patterns using statistical methods to detect exploits and anomalies; balanced detection accuracy while minimizing false-positive rates
- Enabled team scaling by establishing Markdown documentation framework with Git version control; streamlined onboarding process for distributed contributors across multiple time zones

PROJECTS

games.random - AI-Powered Game Development Platform

2024 – Present

- Achieved 85%+ syntactically correct code generation by engineering few-shot learning prompt templates for Claude API; enables non-programmers to create functional JavaScript games from natural language descriptions
- Reduced runtime errors by 40% by building real-time validation pipeline with Monaco editor; provides instant syntax highlighting and error feedback for AI-generated code through immediate feedback loops
- Open-sourced on GitHub with comprehensive documentation of LLM prompt engineering techniques, API integration patterns, and code generation evaluation methods to advance research in AI-assisted creative coding